RepoChat-TestCases (TC-001 to TC-131)

Query handling (27 cases)

ID	Title	Level	Priority	Preconditions	Expected result	Actual result	Status	Automated in
TC-001	Full retrieval pipeline returns generated answer for a relevant query	Unit	High	Repository ingested	Generated answer is returned	As expected	Pass	queryController.test.ts
TC-002	Irrelevant query is blocked before the pipeline runs	Unit	High	Repository ingested	Query rejected, pipeline not run	As expected	Pass	queryController.test.ts
TC-003	Cached response returned immediately on a semantic cache match	Unit	High	Matching cache entry exists	Cached response returned, pipeline skipped	As expected	Pass	queryController.test.ts
TC-004	Fallback message when embedding provider returns no embedding	Unit	High	Embedding provider returns nothing	Fallback message returned	As expected	Pass	queryController.test.ts
TC-005	Second filter pass when the first pass removes every candidate chunk	Unit	Medium	First filter pass removes all chunks	Second filter pass is attempted	As expected	Pass	queryController.test.ts
TC-006	Repository names returned from the repository service	Unit	Medium	Repository ingested	List of repository names returned	As expected	Pass	queryController.test.ts
TC-007	Repository record returned when it exists, message when it does not	Unit	Medium	Repository ingested	Record if found, otherwise a not-found message	As expected	Pass	queryController.test.ts
TC-008	Query handler forwards user query to controller and sends response	Unit	High	Repository ingested	Response payload sent to the client	As expected	Pass	queryHandler.test.ts
TC-009	Query handler retrieves all repositories and sends standard response	Unit	Medium	Repository ingested	All repositories sent in standard response	As expected	Pass	queryHandler.test.ts
TC-010	Query handler retrieves one repository by URL and sends standard response	Unit	Medium	Repository ingested	One repository sent in standard response	As expected	Pass	queryHandler.test.ts
TC-011	Query pipeline returns generated answer and context stats for a relevant query	Integration	High	Repository ingested	Answer and context stats returned	As expected	Pass	queryPipeline.test.ts
TC-012	Query pipeline rejects irrelevant questions before retrieval runs	Integration	High	Repository ingested	Rejected, retrieval not run	As expected	Pass	queryPipeline.test.ts
TC-013	Query pipeline returns cached answer without retrieval or generation	Integration	High	Cached answer exists	Cached answer returned, no retrieval or generation	As expected	Pass	queryPipeline.test.ts
TC-014	Relevance guard is called with the exact query it was given	Unit	Medium	Repository ingested	Guard receives the unmodified query	As expected	Pass	guards.test.ts
TC-015	Guard returns the relevant result unchanged	Unit	Medium	Repository ingested	Relevant result passed through	As expected	Pass	guards.test.ts
TC-016	Guard returns the irrelevant result unchanged	Unit	Medium	Repository ingested	Irrelevant result passed through	As expected	Pass	guards.test.ts
TC-017	Guard propagates a thrown error instead of swallowing it	Unit	Medium	Relevance check throws	Error is propagated	As expected	Pass	guards.test.ts

TC-018	Interpreter calls interpretQuery with the user question before embedding	Unit	Medium	Repository ingested	interpretQuery runs first with the question	As expected	Pass	queryInterpreter.test.ts
TC-019	Interpreter returns embedding, filters, and hypothetical chunk	Unit	Medium	Repository ingested	All three values returned from provider results	As expected	Pass	queryInterpreter.test.ts
TC-020	Interpreter propagates interpretQuery error without wrapping	Unit	Medium	interpretQuery fails	Original error propagated	As expected	Pass	queryInterpreter.test.ts
TC-021	Interpreter propagates createEmbedding error without wrapping	Unit	Medium	createEmbedding fails	Original error propagated	As expected	Pass	queryInterpreter.test.ts
TC-022	Query builder returns a clear empty-context prompt when no chunks	Unit	Medium	No chunks provided	Clear empty-context prompt returned	As expected	Pass	queryBuilder.test.ts
TC-023	Query builder formats each chunk as a labeled code block with the question appended	Unit	Medium	Repository ingested	Labeled code blocks followed by the question	As expected	Pass	queryBuilder.test.ts
TC-024	Query builder builds context stats and preserves file references	Unit	Medium	Repository ingested	Stats built, file references kept	As expected	Pass	queryBuilder.test.ts
TC-025	Query processor builds a simple message list when no chat history	Unit	Medium	No chat history	Simple message list returned	As expected	Pass	queryProcessor.test.ts
TC-026	Query processor prepends system prompt and appends new user message when history exists	Unit	Medium	Chat history exists	System prompt first, new message last	As expected	Pass	queryProcessor.test.ts
TC-027	Query processor propagates provider errors without wrapping	Unit	Medium	Provider fails	Original error propagated	As expected	Pass	queryProcessor.test.ts
Chunk processing and vector search (9 cases)								
ID	Title	Level	Priority	Preconditions	Expected result	Actual Results	Status	Automated in
TC-028	Creates an embedding and stores the chunk with expected metadata	Unit	High		Chunk stored with correct metadata	As expected	Pass	chunkProcessing.test.ts
TC-029	Throws CreationError when embedding provider returns no embedding	Unit	High	Provider returns no embedding	CreationError thrown	As expected	Pass	chunkProcessing.test.ts
TC-030	Logs when a chunk is missing required metadata fields	Unit	Medium	Chunk missing metadata	Missing-field warning logged	As expected	Pass	chunkProcessing.test.ts
TC-031	Throws CreationError when the repository cannot create the chunk	Unit	High	Repository create fails	CreationError thrown	As expected	Pass	chunkProcessing.test.ts
TC-032	Emits progress events while storing chunks and finishes successfully	Unit	Medium		Progress events emitted, then success	As expected	Pass	chunkProcessing.test.ts
TC-033	Emits an error event and rethrows the original OpenAI error on failure	Unit	High	OpenAI error occurs	Error event emitted, original error rethrown	As expected	Pass	chunkProcessing.test.ts
TC-034	Wraps a non-OpenAI error in an AppError and emits an error event	Unit	Medium	Non-OpenAI error occurs	AppError thrown, error event emitted	As expected	Pass	chunkProcessing.test.ts

TC-035	Returns vector search results from the repository	Unit	High	Repository ingested	Search results returned	As expected	Pass	chunkProcessing.test.ts
TC-036	Wraps vector search failures in an AppError	Unit	Medium	Vector search fails	AppError thrown	As expected	Pass	chunkProcessing.test.ts
Retrieval ranking and filtering (15 cases)								
ID	Title	Level	Priority	Preconditions	Expected result	Actual Results	Status	Automated in
TC-037	Reranker returns an empty array when no chunks are provided	Unit	Medium	No chunks	Empty array returned immediately	As expected	Pass	reranker.test.ts
TC-038	Reranker skips reranking and keeps scores when chunk count is at or below topK	Unit	Medium	Chunks at or below topK	No reranking, original scores kept	As expected	Pass	reranker.test.ts
TC-039	Reranker skips reranking when chunk count is at or below the default topK	Unit	Medium	Chunks at or below default topK	No reranking, original scores kept	As expected	Pass	reranker.test.ts
TC-040	Reranker calls scoreChunks with the question and chunks when reranking is needed	Unit	High	Chunks above topK	scoreChunks called with question and chunks	As expected	Pass	reranker.test.ts
TC-041	Reranker returns topK chunks sorted by rerankScore and keeps vectorScore	Unit	High	Repository ingested	Top chunks sorted by rerank score, vector score preserved	As expected	Pass	reranker.test.ts
TC-042	Reranker uses zero for any missing or undefined score	Unit	Medium	Score missing	Score treated as zero	As expected	Pass	reranker.test.ts
TC-043	Reranker propagates scoreChunks errors	Unit	Medium	scoreChunks fails	Error propagated	As expected	Pass	reranker.test.ts
TC-044	Filter drops chunks below the score threshold	Unit	High	Repository ingested	Low-score chunks removed	As expected	Pass	postRetrievalFilter.test.ts
TC-045	Filter applies directory filtering and removes overlapping chunks from the same file	Unit	Medium	Repository ingested	Directory filter applied, overlaps removed	As expected	Pass	postRetrievalFilter.test.ts
TC-046	Filter removes noise files such as README and entrypoint files	Unit	Medium	Repository ingested	Noise files removed	As expected	Pass	postRetrievalFilter.test.ts
TC-047	Filter applies the per-file cap when one file dominates the results	Unit	Medium	One file makes up most results	Per-file cap applied	As expected	Pass	postRetrievalFilter.test.ts
TC-048	Filter logs the filtering stages while processing	Unit	Low	Repository ingested	Stages logged	As expected	Pass	postRetrievalFilter.test.ts
TC-049	Compression is disabled and chunks unchanged when total content is under the threshold	Unit	Medium	Total content under threshold	Chunks unchanged, compression off	As expected	Pass	contextCompression.test.ts
TC-050	Every chunk is compressed when total content exceeds the threshold	Unit	Medium	Total content over threshold	All chunks compressed	As expected	Pass	contextCompression.test.ts
TC-051	Compression provider error is propagated, not swallowed	Unit	Medium	Compression provider fails	Error propagated	As expected	Pass	contextCompression.test.ts
Semantic cache (9 cases)								
ID	Title	Level	Priority	Preconditions	Expected result	Actual Result	Status	Automated in

TC-052	Cache lookup returns null when no matching entry is found	Unit	High	No matching entry	Null returned	As expected	Pass	semanticCache.test.ts
TC-053	Cache returns null when the cached score is below the similarity threshold	Unit	High	Entry score below threshold	Null returned	As expected	Pass	semanticCache.test.ts
TC-054	Cache returns the cached response and context stats when the score meets the threshold	Unit	High	Entry score meets threshold	Cached response and stats returned	As expected	Pass	semanticCache.test.ts
TC-055	Save forwards its arguments to the repository and resolves	Unit	Medium		Arguments forwarded, save resolves	As expected	Pass	semanticCache.test.ts
TC-056	Error from the repository is propagated when cache invalidation fails	Unit	Medium	Invalidation fails	Error propagated	As expected	Pass	semanticCache.test.ts
TC-057	First lookup misses the cache and calls the completion provider	Integration	High	Empty cache	Cache miss, completion provider called	As expected	Pass	semanticCache.test.ts (integration)
TC-058	Repeated identical query hits the cache without calling the completion provider	Integration	High	Query already answered once	Cache hit, provider not called again	As expected	Pass	semanticCache.test.ts (integration)
TC-059	Cache returns null after invalidate is called for that repo URL	Integration	High	Cached entry exists	Null returned after invalidation	As expected	Pass	semanticCache.test.ts (integration)
TC-060	Cached entries are invalidated when re-ingestion occurs for a new SHA	Integration	High	Repo re-ingested with a new SHA	Old cached entries invalidated	As expected	Pass	semanticCache.test.ts (integration)
Ingestion flow (18 cases)								
ID	Title	Level	Priority	Preconditions	Expected result	Actual Result	Status	Automated in
TC-061	Ingest returns early when the repository is already up to date	Unit	Medium	Repo already at latest SHA	Ingestion skipped	As expected	Pass	ingest.test.ts
TC-062	Ingest records an error when cloning does not return a SHA	Unit	High	Clone returns no SHA	Error status recorded	As expected	Pass	ingest.test.ts
TC-063	Ingest aborts and clears the repo directory when storage is over the limit	Unit	High	Storage over limit	Ingestion aborted, directory cleared	As expected	Pass	ingest.test.ts
TC-064	Ingest aborts when the database storage check fails	Unit	High	Storage check fails	Ingestion aborted	As expected	Pass	ingest.test.ts
TC-065	Ingest completes the pipeline for a new repository	Unit	High	New repository	Pipeline completes	As expected	Pass	ingest.test.ts
TC-066	Ingest records an error status when an unexpected exception occurs	Unit	High	Unexpected exception	Error status recorded	As expected	Pass	ingest.test.ts
TC-067	Controller creates ingest status, then calls ingestRepo with the repo URL	Unit	High	Ingestion started	Status created, ingestRepo called with URL	As expected	Pass	ingestController.test.ts
TC-068	Controller logs errors from ingestRepo without throwing them	Unit	Medium	ingestRepo fails	Error logged, not thrown	As expected	Pass	ingestController.test.ts

TC-069	Controller returns the status unchanged when status is not terminal	Unit	Medium	Status in progress	Status returned unchanged	As expected	Pass	ingestController.test.ts
TC-070	Controller deletes ingest progress when status is complete	Unit	Medium	Status complete	Progress deleted	As expected	Pass	ingestController.test.ts
TC-071	Controller deletes ingest progress when status is error	Unit	Medium	Status error	Progress deleted	As expected	Pass	ingestController.test.ts
TC-072	Controller propagates errors from getRepoIngestStatus	Unit	Medium	Status lookup fails	Error propagated	As expected	Pass	ingestController.test.ts
TC-073	Ingest handler starts ingestion and sends the expected 202 response	Unit	High	Ingestion started	202 response sent	As expected	Pass	ingestHandler.test.ts
TC-074	Ingest handler returns the status with a 200 response when a status exists	Unit	Medium	Status exists	200 response with status	As expected	Pass	ingestHandler.test.ts
TC-075	Ingest handler sends a 200 response with null body when no status exists	Unit	Medium	No status exists	200 response with null body	As expected	Pass	ingestHandler.test.ts
TC-076	Pipeline creates a real repo document and chunk documents from fixture files	Integration	High	Fixture files available	Repo and chunk documents created	As expected	Pass	ingestPipeline.test.ts
TC-077	Pipeline records the real ingestion progress sequence in order	Integration	High	Ingestion started	Progress stages recorded in order	As expected	Pass	ingestPipeline.test.ts
TC-078	No duplicate chunks on same-SHA re-ingest; reprocesses on a new SHA	Integration	High	Repo already ingested	No duplicates for same SHA, reprocessed for new SHA	As expected	Pass	ingestPipeline.test.ts
Repository and GitHub access (22 cases)								
ID	Title	Level	Priority	Preconditions	Expected result	Actual Result	Status	Automated in
TC-079	GitHub service reports valid when the remote repository can be reached	Unit	High	Remote reachable	Valid result	As expected	Pass	gitHub.test.ts
TC-080	Private-repository reason returned when remote rejects with an authentication error	Unit	High	Remote rejects authentication	Private-repo reason returned	As expected	Pass	gitHub.test.ts
TC-081	Missing-repository reason returned when the repo does not exist	Unit	High	Remote reports repo missing	Missing-repo reason returned	As expected	Pass	gitHub.test.ts
TC-082	Unreachable reason returned for any other remote error	Unit	Medium	Other remote error	Unreachable reason returned	As expected	Pass	gitHub.test.ts
TC-083	Clones the repository and returns the latest commit hash when valid	Unit	High	Valid repo	Repo cloned, latest SHA returned	As expected	Pass	gitHub.test.ts
TC-084	Throws not-found error when validation fails before cloning	Unit	High	Validation fails	Not-found error thrown	As expected	Pass	gitHub.test.ts
TC-085	Returns the SHA for the requested branch when the remote exposes it	Unit	Medium	Branch exists	Branch SHA returned	As expected	Pass	gitHub.test.ts

TC-086	Throws not-found error when the requested branch is missing	Unit	Medium	Branch missing	Not-found error thrown	As expected	Pass	gitHub.test.ts
TC-087	Creates a new repository record when no existing repo is found	Unit	High	No existing repo	New record created	As expected	Pass	repoProcessing.test.ts
TC-088	Updates an existing repository instead of creating a duplicate	Unit	High	Repo already exists	Existing record updated	As expected	Pass	repoProcessing.test.ts
TC-089	Updates repo metadata and clears chunks for the repo	Unit	Medium	Repo already exists	Metadata updated, chunks cleared	As expected	Pass	repoProcessing.test.ts
TC-090	Returns the result from updating the latest SHA	Unit	Medium	Repo already exists	Update result returned	As expected	Pass	repoProcessing.test.ts
TC-091	Returns the result from updating the last-accessed timestamp	Unit	Low	Repo already exists	Update result returned	As expected	Pass	repoProcessing.test.ts
TC-092	Returns the repo found by URL from the repository layer	Unit	Medium	Repo already exists	Repo returned	As expected	Pass	repoProcessing.test.ts
TC-093	Passes the file tree update through to the repository layer	Unit	Medium	Repo already exists	File tree update forwarded	As expected	Pass	repoProcessing.test.ts
TC-094	Returns all repositories from the repository layer	Unit	Medium		All repositories returned	As expected	Pass	repoProcessing.test.ts
TC-095	Deletes the repo and its chunks together	Unit	High	Repo and chunks exist	Repo and chunks deleted	As expected	Pass	repoProcessing.test.ts
TC-096	Creates the configured storage directory when it does not exist	Unit	Medium	Directory missing	Directory created	As expected	Pass	files.test.ts
TC-097	Deletes all nested files and folders inside the target directory	Unit	Medium		Directory contents deleted	As expected	Pass	files.test.ts
TC-098	Collects parseable files, skipping ignored folders and unsupported extensions	Unit	High	Repo Ingested	Only parseable files collected	As expected	Pass	files.test.ts
TC-099	Builds a file tree that excludes ignored folders and unsupported files	Unit	Medium	Repo Ingested	File tree excludes ignored and unsupported items	As expected	Pass	files.test.ts
TC-100	Throws a not-found error when the root directory does not exist	Unit	Medium	Root directory missing	Not-found error thrown	As expected	Pass	files.test.ts
Ingest progress tracking (10 cases)								
ID	Title	Level	Priority	Preconditions	Expected result	Actual Result	Status	Automated in
TC-101	Forwards the repo URL to createIngestProgress and returns its result	Unit	Medium	Repo exists	URL forwarded, result returned	As expected	Pass	ingestProgress.test.ts
TC-102	Propagates errors from createIngestProgress	Unit	Medium	createIngestProgress fails	Error propagated	As expected	Pass	ingestProgress.test.ts
TC-103	Returns the status object from getIngestProgressStatus	Unit	Medium	Ingest successful	Status object returned	As expected	Pass	ingestProgress.test.ts
TC-104	Calls deleteIngestProgress and resolves	Unit	Medium	Ingest Started	Delete called, resolves	As expected	Pass	ingestProgress.test.ts
TC-105	Propagates errors from deleteIngestProgress	Unit	Medium	deleteIngestProgress fails	Error propagated	As expected	Pass	ingestProgress.test.ts

TC-106	Records one history entry per stage and keeps them in stage order	Integration	High	Ingest Started	One entry per stage, in order	As expected	Pass	ingestProgress.test.ts (integration)
TC-107	Replaces the existing history entry when the same stage is updated twice	Integration	Medium	Stage already recorded	Entry replaced, not duplicated	As expected	Pass	ingestProgress.test.ts (integration)
TC-108	Stores a single error history entry for a terminal failure	Integration	High	Terminal failure occurs	One error entry stored	As expected	Pass	ingestProgress.test.ts (integration)
TC-109	Returns the latest status snapshot when read mid-sequence	Integration	Medium	Ingestion mid-sequence	Latest snapshot returned	As expected	Pass	ingestProgress.test.ts (integration)
TC-110	Removes the ingest progress document from the database	Integration	Medium	Progress document exists	Document removed	As expected	Pass	ingestProgress.test.ts (integration)
Storage management (11 cases)								
ID	Title	Level	Priority	Preconditions	Expected result	Actual Result	Status	Automated in
TC-111	Calculates estimated chunks and storage totals	Unit	Medium	Ingest started	Correct estimates and totals	As expected	Pass	storage.test.ts
TC-112	Returns allowed true when storage is below the limit without the safe buffer	Unit	High	Storage below limit	Allowed: true	As expected	Pass	storage.test.ts
TC-113	Returns allowed false when the safe buffer exceeds the given limit	Unit	High	Safe buffer exceeds limit	Allowed: false	As expected	Pass	storage.test.ts
TC-114	Returns allowed true when projected usage is below the critical threshold	Unit	High	Projected usage below critical threshold	Allowed: true	As expected	Pass	storage.test.ts
TC-115	Deletes oldest repositories and compacts the database when usage is critically high	Unit	High	Usage critically high	Oldest repos deleted, database compacted	As expected	Pass	storage.test.ts
TC-116	Returns allowed false when deleting repositories cannot free enough space	Unit	High	Deletion cannot free enough space	Allowed: false	As expected	Pass	storage.test.ts
TC-117	Returns mapped database stats fields from getDatabaseStorageStats	Unit	Low	Ingest successful	Fields mapped correctly	As expected	Pass	storage.test.ts
TC-118	Deletes the oldest real repo when storage is over the critical threshold	Integration	High	Storage over critical threshold	Oldest repo deleted	As expected	Pass	storage.test.ts (integration)
TC-119	Invalidates the semantic cache for each deleted repo while freeing space	Integration	High	Repos being deleted	Cache invalidated per deleted repo	As expected	Pass	storage.test.ts (integration)
TC-120	Returns not allowed when the database is empty and storage is still critical	Integration	High	Empty database, storage critical	Not allowed	As expected	Pass	storage.test.ts (integration)
TC-121	Calls compactDatabase once after the deletion loop finishes	Integration	Medium	DB at capacity	compactDatabase called exactly once	As expected	Pass	storage.test.ts (integration)
Smoke and infrastructure (5 cases)								
ID	Title	Level	Priority	Preconditions	Expected result	Actual Result	Status	Automated in

TC-122	Mongoose is connected to the in-memory server	Integration	Low		Connection established	As expected	Pass	dbSmoke.test.ts
TC-123	Can write and read a document	Integration	Low		Document written and read back	As expected	Pass	dbSmoke.test.ts
TC-124	Jest is working	Unit	Low		Test runs and passes	As expected	Pass	smoke.test.ts
TC-125	TypeScript types work	Unit	Low		Typed test runs and passes	As expected	Pass	smoke.test.ts
TC-126	Playwright is working and the frontend is reachable	End-to-end	Medium	Frontend running	Page loads	As expected	Pass	smoke.spec.ts

End-to-end user flows (5 cases)

ID	Title	Level	Priority	Preconditions	Expected result	Actual Result	Status	Automated in
TC-127	Shows loading progress for a valid repo and navigates to /chat when ingestion completes	End-to-end	High	Frontend and backend running	Progress shown during ingestion, then navigates to /chat	As expected	Pass	ingestion.spec.ts
TC-128	Shows an error for an invalid repo and does not navigate	End-to-end	High	Frontend and backend running	Error shown, stays on the homepage	As expected	Pass	ingestion.spec.ts
TC-129	Shows a visible answer and referenced files for a relevant question	End-to-end	High	Repo ingested, on the chat page	Answer and referenced files visible	As expected	Pass	query.spec.ts
TC-130	Shows a rejection message for an irrelevant question	End-to-end	High	Repo ingested, on the chat page	Rejection message shown	As expected	Pass	query.spec.ts
TC-131	Keeps both turns in chat history for two questions in the same session	End-to-end	High	Repo ingested, on the chat page	Both questions and answers remain in chat history	As expected	Pass	query.spec.ts